

Note: Coupled channel formalism: CCFULL

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1 Coupled-channels approach

1.1 Coupled-channels equation

In heavy-ion fusion reactions, the collision process is inherently dynamic, involving strong interplay between the relative motion and the intrinsic degrees of freedom of the colliding nuclei. The total fusion probability is significantly affected by such couplings, especially at energies near and below the Coulomb barrier. Consequently, it is essential for a quantitative description of fusion to accurately incorporate the channel coupling effects. This necessitates employing a reaction Hamiltonian that explicitly accounts for these internal structures. Such Hamiltonian can be generally written as

$$H = -\frac{\hbar^2}{2\mu}\nabla^2 + V^{(0)}(r) + H_{\text{int}}(\xi) + V_{\text{coup}}(r, \xi), \quad (1)$$

where r denotes the coordinate of the relative motion between the target and the projectile nuclei, and μ is the reduced mass of the reaction system. $V^{(0)}(r)$ is the bare potential, which is the sum of the Coulomb and the nuclear potentials. $H_{\text{int}}(\xi)$ describes the intrinsic Hamiltonian that governs the excitation spectrum of the colliding nuclei, where ξ represents the internal degrees of freedom. Finally, $V_{\text{coup}}(r, \xi)$ represents the coupling potential between these intrinsic excitations and the relative motion. In conventional coupled-channels calculations, the coupling potential V_{coup} is typically constructed based on the geometrical collective model.

For each total angular momentum J , the total wave function $\Psi(r, \xi)$ is expanded as

$$\Psi(r, \xi) = \sum_n \frac{\phi_n(r)}{r} |n(\xi)\rangle, \quad (2)$$

where $|n(\xi)\rangle$ is the eigen-functions of the intrinsic Hamiltonian $H_{\text{int}}(\xi)$, satisfying $H_{\text{int}}(\xi)|n(\xi)\rangle = \epsilon_n |n(\xi)\rangle$. Here ϵ_n is the eigen-energy of the state $|n(\xi)\rangle$. Latter, I denote the $|n(\xi)\rangle$ as $|n\rangle$. For simplicity, we have used the iso-centrifugal approximation and ignored the spin of the intrinsic state $|n\rangle$. Some reference have proofed these approximations are good approximation, i recommend you can read these articles [Tan87]. The coupled-channels equations for $\phi_n(r)$ are obtained as

$$\begin{aligned} \left[-\frac{\hbar^2}{2\mu} \frac{d^2}{dr^2} + \frac{J(J+1)\hbar^2}{2\mu r^2} + V^{(0)}(r) - E + \epsilon_n \right] \phi_n(r) &= - \sum_m \langle n | V_{\text{coup}}(r) | m \rangle \phi_m(r) \\ &\equiv - \sum_m V_{nm}(r) \phi_m(r), \end{aligned} \quad (3)$$

where E is the incident energy in the center of mass frame. The coupled-channels equations are solved by imposing the incoming wave boundary condition (IWBC), that is, there are only incoming waves at $r \leq r_{\text{min}}$, for which r_{min} is taken to be the minimum position of the Coulomb pocket inside the barrier. In addition, one imposes the boundary condition that there are only the outgoing waves at infinity for all the channels except for the entrance channel ($n = 1$), which has the incoming wave as well.

The flux-normalized boundary conditions are

$$\begin{aligned} \phi_n(r) &\longrightarrow \delta_{n1} H_J^{(-)}(k_n r) + R_n^{(J)} H_J^{(+)}(k_n r), \quad r \rightarrow \infty, \\ \phi_n(r) &\longrightarrow T_n^{(J)} \exp \left[-i \int_{r_{\text{min}}}^r k_n(r') dr' \right], \quad r \rightarrow r_{\text{min}}^+, \end{aligned} \quad (4)$$

where the entrance channel is $n = 1$ and

$$k_n(r) = \left\{ \frac{2\mu}{\hbar^2} \left[E - \epsilon_n - \frac{J(J+1)\hbar^2}{2\mu r^2} - V_{nn}^{\text{eff}}(r) \right] \right\}^{1/2}, \quad (5)$$

$$k_n = \lim_{r \rightarrow \infty} k_n(r), \quad k \equiv k_1. \quad (6)$$

The integral in Eq. (4) is required when the local wave number varies with radius; the frequently used $\exp[-ik_n(r_{\text{min}})(r - r_{\text{min}})]$ is its local constant- k approximation. Current conservation gives

$$P_J(E) = \sum_{n \in \text{open}} \frac{k_n(r_{\text{min}})}{k} |T_n^{(J)}|^2 = 1 - \sum_{n \in \text{open}} \frac{k_n}{k} |R_n^{(J)}|^2, \quad (7)$$

and therefore

$$\sigma_{\text{fus}}(E) = \frac{\pi}{k^2} \sum_{J=0}^{\infty} (2J+1) P_J(E) = \frac{\pi \hbar^2}{2\mu E} \sum_{J=0}^{\infty} (2J+1) P_J(E). \quad (8)$$

Closed asymptotic channels are omitted from the reflected-flux sum but may remain in the coupled equations.

1.2 Coupled potential matrix

The channel coupling is separated into nuclear and Coulomb parts,

$$V_{nm}(r) = V_{nm}^{(N)}(r) + V_{nm}^{(C)}(r). \quad (9)$$

For an axially symmetric target, the surface in its body-fixed frame is

$$R_T(\theta) = R_T c(\boldsymbol{\beta}) \left[1 + \sum_{\lambda \geq 2} \beta_\lambda Y_{\lambda 0}(\theta) \right], \quad (10)$$

where $R_T = r_{\text{coup}} A_T^{1/3}$ and $c(\boldsymbol{\beta})$ may be chosen to conserve the nuclear volume. The present CCFULL input retains $\lambda = 2$ and 4. A static octupole extension adds $\lambda = 3$. Here octupole means multipolarity $\lambda = 3$; it does not mean $\lambda = 8$ or an eight-level truncation.

1.2.1 Rotational coupling of the nuclear potential

Let

$$R_0 = r_0 \left(A_P^{1/3} + A_T^{1/3} \right), \quad (11)$$

$$\hat{O} = R_T [\beta_2 Y_{20}(\theta) + \beta_3 Y_{30}(\theta) + \beta_4 Y_{40}(\theta)]. \quad (12)$$

The all-orders nuclear interaction is obtained by shifting the target surface inside the Woods–Saxon function,

$$V_N(r, \hat{O}) = - \frac{V_0}{1 + \exp[(r - R_0 - \hat{O})/a_0]}, \quad V_N^{(0)}(r) = V_N(r, 0). \quad (13)$$

Thus the nonlinear dependence on the deformation is retained rather than truncated in powers of β_λ .

For the rotating-frame or no-Coriolis basis $|I0\rangle$, the Gaunt integral gives

$$\langle I0 | Y_{\lambda 0} | I'0 \rangle = \sqrt{\frac{(2I+1)(2\lambda+1)(2I'+1)}{4\pi}} \begin{pmatrix} I & \lambda & I' \\ 0 & 0 & 0 \end{pmatrix}^2. \quad (14)$$

Equation (14) follows from

$$\int Y_{I0}^* Y_{\lambda 0} Y_{I'0} d\Omega = \sqrt{\frac{(2I+1)(2\lambda+1)(2I'+1)}{4\pi}} \begin{pmatrix} I & \lambda & I' \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} I & \lambda & I' \\ 0 & 0 & 0 \end{pmatrix}. \quad (15)$$

Consequently,

$$O_{II'} = R_T \sum_{\lambda=2}^4 \beta_\lambda \sqrt{\frac{(2I+1)(2\lambda+1)(2I'+1)}{4\pi}} \begin{pmatrix} I & \lambda & I' \\ 0 & 0 & 0 \end{pmatrix}^2, \quad (16)$$

where the sum contains $\lambda = 2, 3, 4$ when octupole rotation is enabled.

Diagonalize this finite intrinsic-space matrix,

$$\hat{O}|\alpha\rangle = \lambda_\alpha |\alpha\rangle, \quad c_{\alpha I} \equiv \langle \alpha | I0 \rangle. \quad (17)$$

Because a function of $\hat{O}$ is diagonal in the same eigenbasis,

$$\langle I0 | V_N(r, \hat{O}) | I'0 \rangle = \sum_{\alpha} c_{\alpha I}^* c_{\alpha I'} V_N(r, \lambda_\alpha), \quad (18)$$

$$V_{II'}^{(N)}(r) = \sum_{\alpha} c_{\alpha I}^* c_{\alpha I'} V_N(r, \lambda_\alpha) - V_N^{(0)}(r) \delta_{II'}. \quad (19)$$

The subtraction prevents double counting because $V_N^{(0)}$ already occurs in the bare potential.

The $3j$ symbol in Eq. (14) enforces

$$|I - I'| \leq \lambda \leq I + I', \quad I + I' + \lambda \text{ even}. \quad (20)$$

For $\lambda = 3$, $I + I'$ is odd. The octupole term therefore connects opposite-parity members of an alternating-parity or parity-doublet band. A static β_3 rotor requires both parities in the channel basis; adding only the number β_3 to a positive-parity ground-band basis would not be a consistent Hamiltonian extension.

1.2.2 Octupole and triaxial rotor Hamiltonian

The intrinsic quadrupole deformation is parameterized by

$$\alpha_{20} = \beta_2 \cos \gamma, \quad \alpha_{22} = \alpha_{2,-2} = \frac{\beta_2 \sin \gamma}{\sqrt{2}}, \quad 0 \leq \gamma \leq 60^\circ. \quad (21)$$

The laboratory-frame quadrupole operator used by the code is consequently

$$\hat{Q}_2 = \beta_2 \cos \gamma D_{00}^2 + \frac{\beta_2 \sin \gamma}{\sqrt{2}} (D_{02}^2 + D_{0,-2}^2). \quad (22)$$

Equation (22) reduces continuously to the axial $K = 0$ operator when $\gamma = 0$. The asymmetric-rotor intrinsic Hamiltonian is written as

$$\hat{H}_{\text{rot}} = A_1 \hat{I}_1^2 + A_2 \hat{I}_2^2 + A_3 \hat{I}_3^2, \quad A_k = C \left[\sin^2 \left(\gamma - \frac{2\pi k}{3} \right) \right]^{-1}, \quad (23)$$

where the common scale C is fixed so that the axial $K = 0, 2^+$ energy approaches the input value E_{2T} . This is the irrotational Davydov–Filippov form [DF58]. For each I , the Hamiltonian is diagonalized in the even- K , D_2 -symmetric space. The implemented low-energy truncation retains the lowest dominant $K \simeq 0$ state and, for $I \geq 2$, the lowest dominant $|K| \simeq 2$ state. Therefore $I_{\text{max}} = 6$ gives the fixed target-channel count

$$N_T = 1 (I = 0) + 1 (I = 1) + 2 \sum_{I=2}^6 1 = 12. \quad (24)$$

A fixed count is essential because all Numerov snapshots in a POD basis must belong to the same vector space.

The reflection-asymmetric intrinsic terms remain

$$\hat{O}_T = R_T \left[\hat{Q}_2 + \beta_3 Y_{30} + \beta_4 Y_{40} \right]. \quad (25)$$

The coupled-channel Hamiltonian used by the physical solver is then

$$\hat{H}^{(J)}(r) = \left[-\frac{\hbar^2}{2\mu} \frac{d^2}{dr^2} + \frac{J(J+1)\hbar^2}{2\mu r^2} + V^{(0)}(r) \right] \mathbf{1} + \hat{H}_{\text{rot}} + \hat{V}_N^{\text{coup}}(r, \hat{O}_T) + \hat{V}_C^{\text{coup}}(r), \quad (26)$$

where $\hat{V}_N^{\text{coup}} = V_N(r, \hat{O}_T) - V_N(r, 0)\mathbf{1}$. Thus γ enters both the intrinsic spectrum through Eq. (23) and the quadrupole coupling matrix through Eq. (22). The β_3 and β_4 components are axial in the present version. Odd-spin channels have parity $(-1)^I$ and receive the input offset ΔE_{odd} . Consequently, β_3 connects the opposite-parity parts of the alternating-parity band, while the triaxial quadrupole term mixes K by $\Delta K = \pm 2$. The low-lying γ -band sensitivity found in fusion barrier distributions by Rumin, Hagino, and Takigawa [RHT99] motivates retaining the $|K| \simeq 2$ band. Their one-phonon γ -band calculation and the present static triaxial rotor are related physical tests but are not identical models.

For ^{154}Sm , the value $\gamma = 5^\circ$ used in the supplied input is consistent with the recent shape extraction $\beta = 0.2925(25)$ and $\gamma = 5.0(15)^\circ$ from the IVGDR γ -decay analysis of Kleemann *et al.* [KPFG⁺25]. This external value is a physically motivated center point; it is not imposed as a fusion-data result.

1.2.3 Rotational coupling of the Coulomb potential

For r outside a localized target charge distribution,

$$\frac{1}{|\mathbf{r} - \mathbf{r}'|} = \sum_{\lambda\mu} \frac{4\pi}{2\lambda+1} \frac{r'^\lambda}{r^{\lambda+1}} Y_{\lambda\mu}^*(\hat{\mathbf{r}}) Y_{\lambda\mu}(\hat{\mathbf{r}}'). \quad (27)$$

With

$$Q_{\lambda\mu} = \int d^3 r' \rho_T(\mathbf{r}') r'^\lambda Y_{\lambda\mu}(\hat{\mathbf{r}}'), \quad (28)$$

the projectile–target Coulomb interaction becomes

$$V_C(\mathbf{r}) = \frac{Z_P Z_T e^2}{r} + Z_P e^2 \sum_{\lambda \geq 2, \mu} \frac{4\pi}{2\lambda + 1} \frac{Q_{\lambda\mu}}{r^{\lambda+1}} Y_{\lambda\mu}^*(\hat{\mathbf{r}}). \quad (29)$$

For a sharp, uniformly charged axial surface, expansion to first order in β_λ yields

$$V_{\text{coup}}^{(C,1)}(r, \theta) = \sum_{\lambda \geq 2} \frac{3Z_P Z_T e^2}{2\lambda + 1} \frac{R_T^\lambda}{r^{\lambda+1}} \beta_\lambda Y_{\lambda 0}(\theta). \quad (30)$$

The corresponding rotor matrix element is therefore

$$V_{II'}^{(C,1)}(r) = \sum_{\lambda \geq 2} \frac{3Z_P Z_T e^2}{2\lambda + 1} \frac{R_T^\lambda}{r^{\lambda+1}} \beta_\lambda \sqrt{\frac{(2I+1)(2\lambda+1)(2I'+1)}{4\pi}} \begin{pmatrix} I & \lambda & I' \\ 0 & 0 & 0 \end{pmatrix}^2. \quad (31)$$

In particular, the added octupole rotational term is

$$V_{II'}^{(C,\lambda=3)}(r) = \frac{3Z_P Z_T e^2}{7} \frac{R_T^3}{r^4} \beta_3 \sqrt{\frac{7(2I+1)(2I'+1)}{4\pi}} \begin{pmatrix} I & 3 & I' \\ 0 & 0 & 0 \end{pmatrix}^2. \quad (32)$$

For comparison with the standard CCFULL rotor, keeping second order in β_2 and first order in β_4 gives

$$\begin{aligned} V_{II'}^{(C;2,4)}(r) &= \frac{3Z_P Z_T e^2}{5} \frac{R_T^2}{r^3} \sqrt{\frac{5(2I+1)(2I'+1)}{4\pi}} \left(\beta_2 + \frac{2}{7} \sqrt{\frac{5}{\pi}} \beta_2^2 \right) \begin{pmatrix} I & 2 & I' \\ 0 & 0 & 0 \end{pmatrix}^2 \\ &+ \frac{3Z_P Z_T e^2}{9} \frac{R_T^4}{r^5} \sqrt{\frac{9(2I+1)(2I'+1)}{4\pi}} \left(\beta_4 + \frac{9}{7\sqrt{\pi}} \beta_2^2 \right) \begin{pmatrix} I & 4 & I' \\ 0 & 0 & 0 \end{pmatrix}^2. \end{aligned} \quad (33)$$

Equations (32) and (33) define a controlled convention: first order in β_3 , second order in β_2 , and first order in β_4 . Terms such as $\beta_2\beta_3$ or β_3^2 belong to a higher-order extension and must be added consistently to both the shape expansion and the Coulomb multipoles.

2 Numerical solution methods

2.1 The modified Numerov method

Introduce the channel vector $\phi = (\phi_1, \dots, \phi_{N_{\text{ch}}})^T$. Equation (3) can be written as

$$\phi''(r) = \mathbf{A}(r)\phi(r), \quad (34)$$

with

$$A_{nn'}(r) = \frac{2\mu}{\hbar^2} \left[\left(\frac{J(J+1)\hbar^2}{2\mu r^2} + V^{(0)}(r) + \epsilon_n - E \right) \delta_{nn'} + V_{nn'}(r) \right]. \quad (35)$$

The positive sign of the off-diagonal term follows by moving the right-hand side of Eq. (3) to the left before isolating ϕ'' .

For a uniform mesh $r_i = r_{\text{min}} + (i-1)h$, Taylor expansion about r_i gives

$$\phi_{i+1} - 2\phi_i + \phi_{i-1} = \frac{h^2}{12} (\phi_{i+1}'' + 10\phi_i'' + \phi_{i-1}'') + \mathcal{O}(h^6). \quad (36)$$

Substituting Eq. (34) and defining

$$\mathbf{F}_i = \mathbf{1} - \frac{h^2}{12} \mathbf{A}_i \quad (37)$$

yields the coupled Numerov recurrence

$$\mathbf{F}_{i+1}\phi_{i+1} = \left(2\mathbf{1} + \frac{5h^2}{6} \mathbf{A}_i \right) \phi_i - \mathbf{F}_{i-1}\phi_{i-1} + \mathcal{O}(h^6). \quad (38)$$

CCFULL uses an algebraically stabilized modified form. With

$$\chi_i = \mathbf{F}_i \phi_i, \quad (39)$$

the recurrence used in the program and in the Numerov-DBM operator is

$$\chi_{i+1} + \chi_{i-1} - \left[\left(\frac{h^2}{\sqrt{12}} \mathbf{A}_i + \sqrt{3} \mathbf{1} \right)^2 - \mathbf{1} \right] \chi_i = 0. \quad (40)$$

Matrix multiplication is intended in Eq. (40); the recurrence is not applied separately to each $A_{nn'}$ entry.

For each independent inner channel s , the initial conditions are

$$\phi_n^{(s)}(r_{\min}) = \delta_{ns}, \quad \left. \frac{d\phi_n^{(s)}}{dr} \right|_{r_{\min}} = -ik_s(r_{\min})\delta_{ns}. \quad (41)$$

Runge-Kutta supplies the second mesh point and Eq. (40) propagates the fundamental solution matrix outward. Before Coulomb matching, $\phi_i = \mathbf{F}_i^{-1} \chi_i$ is reconstructed.

At $r_{\max}$ each propagated column is decomposed as

$$\phi_n^{(s)}(r) = C_{ns} H_J^{(-)}(k_n r) + D_{ns} H_J^{(+)}(k_n r). \quad (42)$$

The two coefficients are obtained from the wavefunction and derivative (or equivalently two neighboring mesh values) by the 2×2 solve

$$\begin{pmatrix} H_n^- & H_n^+ \\ H_n^{-'} & H_n^{+'} \end{pmatrix} \begin{pmatrix} C_{ns} \\ D_{ns} \end{pmatrix} = \begin{pmatrix} \phi_n^{(s)} \\ \phi_n^{(s)'} \end{pmatrix}_{r=r_{\max}}. \quad (43)$$

The physical solution $\phi = \sum_s T_s \phi^{(s)}$ must contain one incoming wave in the entrance channel and none in the others, hence

$$\mathbf{CT} = \mathbf{e}_1, \quad \mathbf{T} = \mathbf{C}^{-1} \mathbf{e}_1, \quad \mathbf{R} = \mathbf{DT}. \quad (44)$$

Substitution in Eq. (7) completes the calculation of $P_J(E)$ and Eq. (8) gives the fusion cross section.

2.2 The discrete basis method

On the same mesh, the three-point approximation

$$-\frac{\hbar^2}{2\mu} \phi_n''(r_i) \simeq -t (\phi_{n,i+1} - 2\phi_{n,i} + \phi_{n,i-1}), \quad t = \frac{\hbar^2}{2\mu h^2}, \quad (45)$$

turns the coupled equations into

$$\sum_{n'i'} (H_{ni,n'i'} - E \delta_{nn'} \delta_{ii'}) \phi_{n'i'} = 0, \quad (46)$$

where

$$\begin{aligned} H_{ni,n'i'} &= \delta_{nn'} (-t \delta_{i,i'+1} + 2t \delta_{ii'} - t \delta_{i,i'-1}) \\ &+ \delta_{nn'} \delta_{ii'} \left[\frac{J(J+1)\hbar^2}{2\mu r_i^2} + V^{(0)}(r_i) + \epsilon_n \right] + \delta_{ii'} V_{nn'}(r_i). \end{aligned} \quad (47)$$

The radial hopping $-t$ connects (n, i) with $(n, i \pm 1)$ only; it is not an off-diagonal channel coupling. Channel mixing at fixed i comes from $V_{nn'}(r_i)$.

The inner ghost point follows from the incoming-wave condition,

$$\phi_{n,0} = e^{ik_n(r_{\min})h} \phi_{n,1}. \quad (48)$$

Eliminating it changes the first diagonal block to

$$\tilde{H}_{ni,n'i'} = H_{ni,n'i'} - E\delta_{nn'}\delta_{ii'} - te^{ik_n(r_{\min})h}\delta_{nn'}\delta_{ii'}\delta_{i1}. \quad (49)$$

To impose the outer condition, introduce interior coordinate vectors ψ_{sj} and outgoing channel vectors $\psi_{s,+}$:

$$\psi_{sj}(n'i') = \delta_{n's}\delta_{i'j}, \quad 1 \leq j \leq N, \quad (50)$$

$$\psi_{s,+}(n'i') = \delta_{n's} \left[\delta_{i',N+1} H_J^{(+)}(k_s r_{N+1}) + \delta_{i',N+2} H_J^{(+)}(k_s r_{N+2}) \right]. \quad (51)$$

The known incident vector $\psi^{(-)}$ has the same two outer entries in channel 1 with $H_J^{(-)}$. The complete scattering vector is

$$\Psi = \sum_{s=1}^{N_{\text{ch}}} \sum_{j=1}^N b_{sj} \psi_{sj} + \sum_{s=1}^{N_{\text{ch}}} R_s \psi_{s,+} + \psi^{(-)}. \quad (52)$$

Collect all unknown coefficients into $\mathbf{x}$. Applying $\tilde{\mathbf{H}}$ gives the square linear system

$$\underbrace{\left[\tilde{\mathbf{H}}\psi_{11}, \dots, \tilde{\mathbf{H}}\psi_{N_{\text{ch}},N}, \tilde{\mathbf{H}}\psi_{1,+}, \dots, \tilde{\mathbf{H}}\psi_{N_{\text{ch}},+} \right]}_{\mathbf{M}} \mathbf{x} = -\tilde{\mathbf{H}}\psi^{(-)}. \quad (53)$$

The first radial coefficients give the inner transmission amplitudes, and the last N_{ch} coefficients give the reflected amplitudes. Their flux-weighted values satisfy Eq. (7); this identity is also a numerical conservation check.

2.3 The modified discrete basis method

The Numerov-DBM retains the boundary-vector construction of Eq. (52) but replaces the three-point residual by Eq. (40). Its interior sparse operator is defined directly by

$$[\mathbf{H}^{(N)}\chi]_i = \chi_{i-1} + \chi_{i+1} - \left[\left(\frac{h^2}{\sqrt{12}} \mathbf{A}_i + \sqrt{3}\mathbf{1} \right)^2 - \mathbf{1} \right] \chi_i. \quad (54)$$

Because $\chi_1 = \mathbf{F}_1\phi_1$, elimination of the inner ghost point adds the boundary block

$$\Delta\tilde{\mathbf{H}}_1^{(N)} = \mathbf{F}_1 \text{diag}\left(e^{ik_n(r_{\min})h}\right). \quad (55)$$

The remaining construction and solution are the same as Eq. (53). This replacement raises the radial accuracy without increasing the number of mesh points and is the Numerov-discretized DBM used in Ref. [LHZ⁺26].

2.4 The R-matrix method

2.4.1 R matrix and S matrix

For one channel, divide space at the channel radius a . The internal Hamiltonian becomes Hermitian on $[0, a]$ after adding the Bloch operator

$$\mathcal{L}(B) = \frac{\hbar^2}{2\mu} \delta(r-a) \left(\frac{d}{dr} - \frac{B}{a} \right). \quad (56)$$

The Bloch-Schrödinger equation is

$$[T + V + \mathcal{L}(B) - E]u(r) = \mathcal{L}(B)u_{\text{ext}}(r). \quad (57)$$

Expand $u_{\text{int}}(r) = \sum_{j=1}^N c_j \varphi_j(r)$ and project with $\langle \varphi_i |$. With

$$C_{ij}(E) = \langle \varphi_i | T + V + \mathcal{L}(B) - E | \varphi_j \rangle, \quad (58)$$

the coefficients satisfy

$$\sum_j C_{ij} c_j = \frac{\hbar^2}{2\mu a} \varphi_i(a) [a u'_{\text{ext}}(a) - B u_{\text{ext}}(a)]. \quad (59)$$

For $B = 0$, substitution back into $u(a) = \sum_i c_i \varphi_i(a)$ yields

$$u(a) = R(E) a u'(a), \quad (60)$$

$$R(E) = \frac{\hbar^2}{2\mu a} \sum_{ij} \varphi_i(a) [\mathbf{C}^{-1}(E)]_{ij} \varphi_j(a). \quad (61)$$

Using the external convention

$$u_{\text{ext}}(r) = H_L^{(-)}(kr) - S_L H_L^{(+)}(kr), \quad (62)$$

and denoting radial derivatives by primes, the boundary relation gives

$$S_L = \frac{H_L^{(-)} - R a H_L^{(-)'} }{H_L^{(+)} - R a H_L^{(+)'} } \Big|_{r=a}. \quad (63)$$

For a one-channel problem with absorption or an inner IWBC, $P_L = 1 - |S_L|^2$. For a real potential with a regular origin and no absorbing boundary, unitarity gives $|S_L| = 1$ and hence zero absorption.

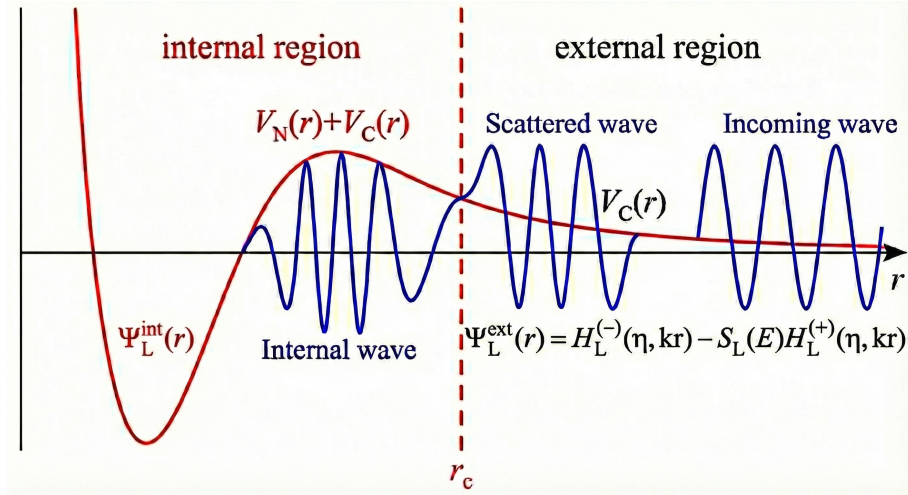


Figure 1: Schematic representation of the R-matrix division of space.

2.4.2 Basis function: Lagrange basis

Let $x_i \in (0, 1)$ be the roots of $P_N(2x_i - 1) = 0$. A regularized Lagrange–Legendre function on $[0, a]$ is

$$\varphi_i(r) = (-1)^{N+i} \left(\frac{r}{ax_i} \right)^\nu \sqrt{ax_i(1-x_i)} \frac{P_N(2r/a-1)}{r-ax_i}, \quad (64)$$

with $\nu = 1$ for a regular radial solution. Its normalization is chosen so that

$$\varphi_i(ax_j) = \frac{\delta_{ij}}{\sqrt{a\lambda_j}}, \quad (65)$$

where λ_j is the Gauss–Legendre weight on $[0, 1]$. Gauss quadrature then gives

$$\langle \varphi_i | \varphi_j \rangle_G = \delta_{ij}, \quad (66)$$

$$\langle \varphi_i | V | \varphi_j \rangle_G = V(ax_i) \delta_{ij}. \quad (67)$$

The kinetic plus Bloch matrix is evaluated analytically, while the local potential is diagonal under Eq. (67).

For an IWBC interval $[r_{\text{abs}}, r_c]$, define $L = r_c - r_{\text{abs}}$ and $x = (r - r_{\text{abs}})/L$. The mapped basis is simply

$$\varphi_i^{[r_{\text{abs}}, r_c]}(r) = L^{-1/2} f_i\left(\frac{r - r_{\text{abs}}}{L}\right), \quad (68)$$

where f_i is the unit-interval form of Eq. (64). Hence potential nodes become $r_i = r_{\text{abs}} + Lx_i$, first-derivative matrices scale as L^{-1} , and kinetic matrices scale as L^{-2} . The two-boundary operator is

$$\mathbf{C} = \langle \varphi_i | T + V + \mathcal{L}_c - \mathcal{L}_{\text{abs}} - E | \varphi_j \rangle. \quad (69)$$

At the inner boundary, the IWBC specifies the logarithmic derivative $u'/u = -ik(r_{\text{abs}})$; at r_c , Eq. (63) supplies the Coulomb matching. This formulation fixes the signs by the outward normals of the two endpoints and avoids treating the inner boundary as a regular origin.

3 Method comparison in CCFULL

3.1 Accuracy

Now, we have introduced several way to solve the coupled-channel equations, and we will compare these method. Specifically, we calculated the sub-barrier fusion reaction $^{16}\text{O} + ^{144}\text{Sm}$. we consider the 3^- vibration motion of ^{144}Sm .

For the calculation parameters,

- The coupling strength $\beta_3 = 0.205$. The excitation energy of the state $3^- = 1.81\text{MeV}$.
- The energy interval ΔE is 0.18 MeV. The space interval Δr is 0.05 fm for the Numerov and DBM method. The basis number is 120 for the R matrix method.
- The Ws potential parameter $V_0 = 105.1\text{MeV}$, $r_0 = 1.1\text{fm}$, $a_0 = 0.75\text{fm}$.

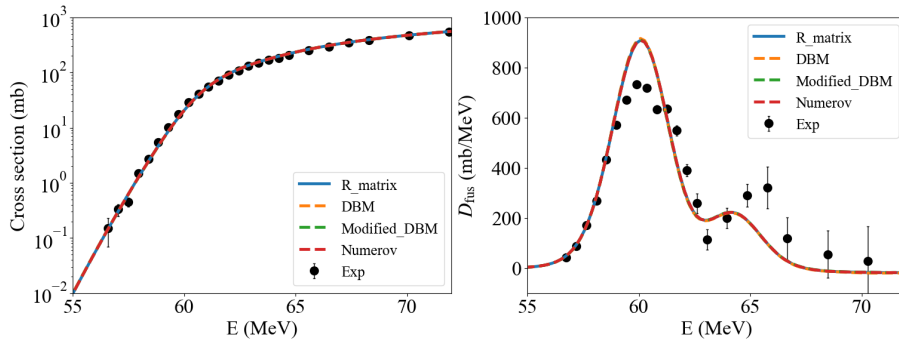


Figure 2: Comparison of experimental fusion cross sections and barrier distribution for the $^{16}\text{O} + ^{144}\text{Sm}$ system with results solved by different methods

3.2 Speed

Here I did a calculation of $^{16}\text{O} + ^{144}\text{Sm}$ system with results solved by different methods, the energy range from 55 to 75 MeV, a total of 20 energy points.

Table 1 reports elapsed times for the stated 20-energy calculation. Numerov is fastest at 1.226 s, followed by the R-matrix calculation at 7.802 s. DBM and modified DBM take about 56 s each because they solve substantially larger linear systems. CPU utilization above 100% indicates threaded numerical-library activity. The 9% Numerov utilization alone is insufficient to identify an I/O bottleneck; profiling would be required for that conclusion.

Table 1: Comparison of program execution time for different numerical methods

Method	CPU utilization	Total Time (s)
Numerov	9%	1.226
Rmatrix	79%	7.802
DBM	106%	56.466
Modified DBM	107%	56.041

3.3 Stability

Next, I calculated the fusion cross-sections of different systems. Specifically, we calculated the sub-barrier fusion reaction ^{16}O , ^{20}Ne , ^{24}Mg , ^{28}Si , ^{40}Ar , ^{40}Ca , ^{58}Ni + ^{144}Sm . we consider the 3^- vibration motion of ^{144}Sm .

For the calculation parameters,

- The coupling strength $\beta_3 = 0.205$. The excitation energy of the state $3^- = 1.81\text{MeV}$.
- The energy interval ΔE is 1 MeV. The space interval Δr is 0.05 fm for the Numerov and DBM method. The basis number is 120 for the R matrix method.
- The W_s potential parameter we adpoted the empirical form is given by:

$$V_N(r) = \frac{-V_0}{1 + \exp[(r - R_P - R_T)/a]}, \quad (70)$$

where

$$\frac{1}{a} = 1.17[1 + 0.53(A_P^{-1/3} + A_T^{-1/3})] \text{ fm}^{-1}, \quad (71)$$

$$r_0 = 1.16 \text{ fm}, \quad (72)$$

$$V_0 = 80 \text{ MeV}, \quad (73)$$

$$R_P = r_0 A_P^{1/3}, \quad R_T = r_0 A_T^{1/3} \quad (74)$$

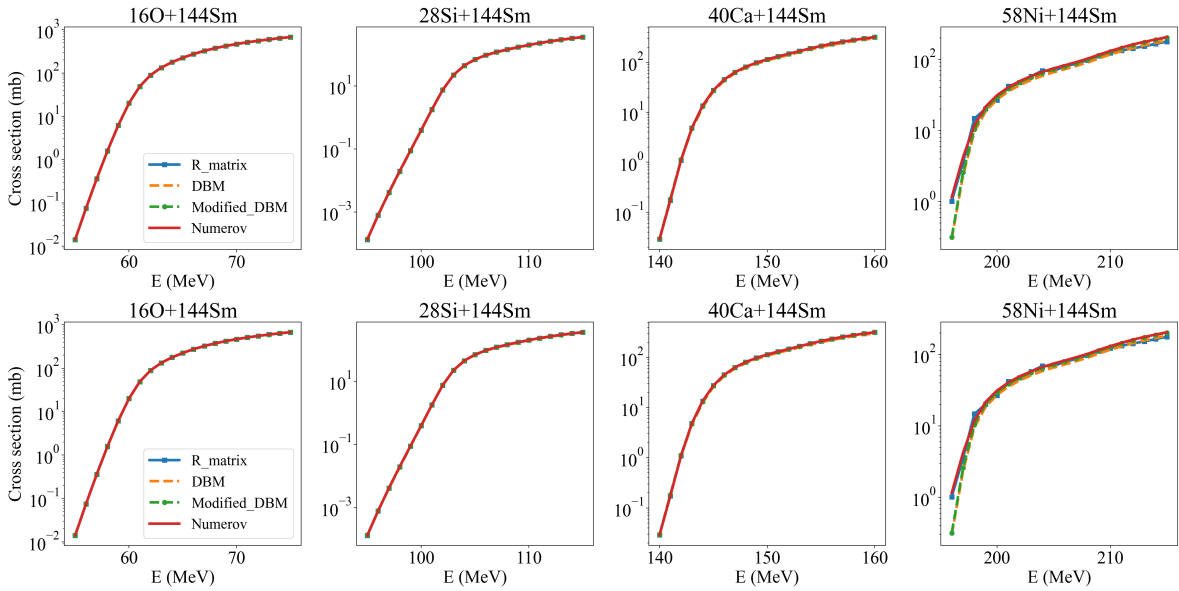


Figure 3: Comparison of fusion cross sections for the ^{16}O , ^{20}Ne , ^{24}Mg , ^{28}Si , ^{40}Ar , ^{40}Ca , ^{58}Ni + ^{144}Sm systems with results solved by different methods

4 Reduced basis method and on-line simulation emulator

4.1 Retained physical models and emulator reference

After the directory consolidation, three physical sources are retained:

1. `Standard Numerov`, the real-potential modified-Numerov calculation;
2. `Hagino original CCFULL`, the independent standard-model baseline; and
3. `Octupole + triaxial  $\gamma$  Numerov`, the physical model used for all new emulator training and validation.

The exact source-directory map is kept in the workbench README so that the long directory names do not obscure the physical discussion here. Every new emulator snapshot, matrix-EIM operator node, response curve, and standalone exact validation point is generated by the third source. Thus a saved emulator and its high-fidelity reference use the same channel Hamiltonian. The standard Numerov and Hagino programs remain physical-model comparisons and are not mixed into the triaxial training set.

4.2 Seven-coordinate training problem

The complete emulator coordinate vector is now

$$\boldsymbol{\theta} = (V_0, r_0, a_0, \beta_2, \beta_3, \beta_4, \gamma). \quad (75)$$

The workbench may activate any non-empty subset of Eq. (75). Every omitted coordinate is recorded as a fixed physical value. At every training or validation point the program performs the following operations:

1. rebuild and diagonalize Eq. (23);
2. transform E2, E3, and E4 matrix elements into the mixed rotor basis;
3. build the all-order Woods–Saxon surface-displacement coupling;
4. solve the coupled radial problem with Numerov and IWBC;
5. phase-align the complex wavefunction snapshot and update the affine POD basis;
6. construct matrix-EIM reduced operator and source nodes; and
7. accumulate partial-wave fusion cross sections and fit the persistent log-cross-section POD/RBF response package.

Custom parameter tables and observable-error adaptive sampling are both available. Exact validation always calls the same triaxial Numerov source at the selected parameter vector and energies. The pointwise high-fidelity error is defined directly by

$$\epsilon_\sigma(E; \boldsymbol{\theta}) = \frac{|\sigma_{\text{emu}}(E; \boldsymbol{\theta}) - \sigma_{\text{tri}}(E; \boldsymbol{\theta})|}{\max(|\sigma_{\text{tri}}(E; \boldsymbol{\theta})|, \sigma_{\text{floor}})}, \quad (76)$$

where “tri” denotes the retained octupole plus triaxial Numerov calculation at exactly the same physical inputs and energy.

4.3 Axial-limit and finite- γ checks

The $\gamma = 0$ input keeps 12 channels so that the emulator dimension is unchanged, but the $|K| = 2$ sector decouples. The resulting 21 cross sections from 55 to 75 MeV reproduce the frozen axial-octupole reference to the printed precision. At 55 MeV,

$$\sigma_{\gamma=0} = 5.73458 \text{ mb}, \quad \sigma_{\gamma=5^\circ} = 5.65066 \text{ mb}. \quad (77)$$

The $\gamma = 5^\circ$ calculation is finite over the complete interval, with $5.65066 \leq \sigma \leq 730.78706$ mb. A controlled speed comparison used the common axial subproblem $\beta_3 = \gamma = 0$, $I_{\max} = 4$, 21 energies from 55 to 75 MeV, and identical reaction, potential, radial-mesh, β_2 , and β_4 inputs. The Standard Numerov and triaxial programs then produced identical printed cross sections, while their median warm run times over seven repetitions were 0.0857 s and 0.5605 s. Thus the fixed triaxial channel space costs a factor of about 6.54 for this matched problem. Within that fixed space, changing only γ from 0° to 5° changed the median time only from 0.5605 s to 0.5696 s; the main cost is the retained channel dimension, not evaluation of the trigonometric γ factors.

A small end-to-end training test has verified the seven-coordinate data path, model metadata, online prediction, and direct triaxial Numerov validation. These checks validate the implementation flow; a production seven-dimensional model still requires a sufficiently dense design and independent errors over the final parameter and energy domain.

4.4 Production calibration tests for $^{16}\text{O}+^{154}\text{Sm}$

Two controlled calibration tests were carried out to document the complete offline–online workflow. Both use the same high-fidelity source, physical inputs, experimental cross sections, energy grid, and numerical tolerances. The only physical difference is whether the triaxial angle is an active coordinate or fixed at its axial value. This construction makes the two tests a direct comparison of the three-dimensional vector $(\beta_2, \beta_4, \gamma)$ and the two-dimensional vector (β_2, β_4) at $\gamma = 0$.

The common physical and numerical inputs are summarized in Table 2. The 39 center-of-mass energies are the complete set in `fusion-exp-8.16.62.154-tb-1995-611.txt`; no experimental energy was discarded or moved to a regular grid. They span $52.4144 \leq E_{\text{cm}} \leq 99.5202$ MeV. The tabulated barrier distribution is read from `fusion-barrier-8.16.62.154-liao-1995.txt`.

Table 2: Common physical and numerical inputs for the two $^{16}\text{O}+^{154}\text{Sm}$ emulator tests.

Quantity	Value
Reaction	$A_P = 16$, $Z_P = 8$, $A_T = 154$, $Z_T = 62$
Radius inputs	$r_{0P} = 1.20$ fm and $r_{0T} = 1.06$ fm
Target rotor	$E_{2+} = 0.082$ MeV, $I_{\max} = 6$, $\beta_3 = 0$, odd-parity offset 0.8 MeV
Additional target band	$E = 1.66$ MeV, coupling 0.11, multipolarity 2
Projectile excitation	$E = 6.13$ MeV, coupling 0.733, multipolarity 3
Nuclear potential	$V_0 = 165$ MeV, $r_0 = 0.95$ fm, $a_0 = 1.05$ fm
Radial calculation	$r_{\max} = 30$ fm, $\Delta r = 0.05$ fm
Partial-wave control	$J_{\max} = 80$, tail window 4, tail tolerance 10^{-4}
Observable grid	all 39 measured cross-section energies from 52.4144 to 99.5202 MeV

For both tests, the candidate parameter vectors are generated with a Latin hypercube in the stated uniform prior box. Adaptive selection begins from a space-filling subset. After fitting a temporary log-response POD/RBF model, the unused candidate with the largest pointwise cross-section error is added. The iteration stops when the largest candidate error is below 1%, or when the maximum of 56 selected points is reached. A “candidate point” here is a complete physical parameter vector; one such point contains the full 39-energy cross-section curve. Eighteen independent parameter vectors are reserved for validation and never enter the adaptive selection.

The matrix-EIM response bank uses 24 wavefunction training snapshots and 36 operator samples, evaluated in four independent energy groups. The response SVD tolerance is 10^{-4} , the cross-section floor is 10^{-10} mb, the partial-wave floor is 10^{-14} mb, and the relative-error denominator uses $\sigma_{\text{floor}} = 10^{-4}$ mb in Eq. (76). Four held-out parameter vectors are additionally recalculated with the standalone triaxial Numerov code at all 39 energies. Table 3 gives the case-specific details.

The two-dimensional selection stopped after 21 points because the candidate error had already crossed the 1% target. Its offline time nevertheless remained close to that of the three-dimensional test because the present implementation first evaluates all 96 candidate responses and then selects the final training subset. A fully sequential active-learning implementation would avoid calculating candidates that are never selected.

Table 3: Training design and validation results. Relative errors are quoted as percentages.

Setting or result	3D: β_2, β_4, γ	2D: β_2, β_4
Fixed coordinates	$\beta_3 = 0$	$\beta_3 = 0, \gamma = 0$
Parameter ranges	$\beta_2 : [0.28, 0.36], \beta_4 : [0, 0.08], \gamma : [0^\circ, 10^\circ]$	$\beta_2 : [0.28, 0.36], \beta_4 : [0, 0.08]$
Random-design seed	20261006	20261008
Candidate / initial / maximum	96 / 20 / 56	96 / 20 / 56
Actually selected points	56	21
Final candidate maximum error	1.825	0.576
Validation parameter points	18	18
Exact Numerov validation points	4	4
Cross-section POD rank	19	10
Partial-wave POD rank	55	20
Independent validation median error	0.138	0.00436
Independent validation maximum error	55.8	0.486
Exact Numerov median error	0.823	0.423
Exact Numerov maximum error	11.6	4.63
Offline wall time	656.8 s	633.1 s

4.4.1 Bayesian calibration and reported uncertainty

The prior is uniform inside each emulator box. For $N = 39$ measured cross sections, the statistical likelihood is

$$p(\mathcal{D}|\boldsymbol{\theta}, s = 1) \propto \exp\left[-\frac{1}{2}\chi^2(\boldsymbol{\theta})\right], \quad \chi^2(\boldsymbol{\theta}) = \sum_{i=1}^N \frac{[\sigma_{\text{emu}}(E_i; \boldsymbol{\theta}) - \sigma_i^{\text{exp}}]^2}{\Delta\sigma_i^2}. \quad (78)$$

For asymmetric experimental errors, the upper or lower uncertainty is chosen according to the sign of the residual. Since both calibrated models have $\chi_\nu^2 \gg 1$, intervals based only on the quoted experimental errors are too narrow. A global error multiplier s is therefore introduced, $\Delta\sigma_i \rightarrow s\Delta\sigma_i$, and marginalized with the scale prior $p(s) \propto 1/s$. Conditional on $\boldsymbol{\theta}$, this gives

$$p(\boldsymbol{\theta}|\mathcal{D}) \propto p(\boldsymbol{\theta})[\chi^2(\boldsymbol{\theta})]^{-N/2}. \quad (79)$$

The reported conservative intervals use Eq. (79). They include a data-driven global error scale but do not constitute a complete model-discrepancy model. Each case uses 400,000 Latin-hypercube importance samples: seed 20261007 for the three-dimensional case and 20261009 for the two-dimensional case.

The fusion-barrier distribution is a posterior diagnostic rather than an additional fit observable. It is evaluated from

$$\begin{aligned} F(E) &\equiv E\sigma_{\text{fus}}(E), \\ D_{\text{fus}}(E) &= \frac{d^2 F(E)}{dE^2} \simeq \frac{F(E + \Delta E) - 2F(E) + F(E - \Delta E)}{(\Delta E)^2}, \\ \Delta E &= 1.8 \text{ MeV}. \end{aligned} \quad (80)$$

All 30 experimental barrier points are shown. Exact Numerov is evaluated for all 30 points. The emulator derivative is available for 26 points; the four lowest centers require $E - \Delta E$ below the trained energy domain.

4.4.2 Three-dimensional deformation case

The three-dimensional maximum-likelihood point inside the trained box is

$$(\beta_2, \beta_4, \gamma)_{\text{best}} = (0.334744, 0.040789, 10.0^\circ). \quad (81)$$

The corresponding scale-marginalized posterior summaries are

$$\beta_2 = 0.33116, \quad 68\% : [0.32471, 0.33755], \quad 95\% : [0.31850, 0.34409], \quad (82)$$

$$\beta_4 = 0.04502, \quad 68\% : [0.03685, 0.05315], \quad 95\% : [0.02841, 0.06080], \quad (83)$$

$$\gamma = 7.64^\circ, \quad 68\% : [1.02^\circ, 9.57^\circ], \quad 95\% : [0.13^\circ, 9.94^\circ]. \quad (84)$$

The weighted correlations are $\rho(\beta_2, \beta_4) = -0.807$, $\rho(\beta_2, \gamma) = 0.229$, and $\rho(\beta_4, \gamma) = -0.293$. The likelihood optimum reaches the upper γ boundary. Thus 10° is a boundary-limited optimum and is not an unconstrained determination of the triaxial angle.

At Eq. (81), the emulator has $\chi^2 = 376.60$ and $\chi_\nu^2 = 10.46$. Recalculation with standalone Numerov gives $\chi^2 = 430.08$ and $\chi_\nu^2 = 11.95$. The median and maximum emulator–Numerov errors on the 39-energy curve are 0.833% and 14.1%, with the maximum at 53.3202 MeV. Figure 4 shows the cross-section calibration, and Fig. 5 shows the barrier diagnostic at the same parameters.

4.4.3 Two-dimensional axial case

With $\gamma = 0$ fixed, the maximum-likelihood point is

$$(\beta_2, \beta_4)_{\text{best}} = (0.330381, 0.044997). \quad (85)$$

The conservative posterior summaries are

$$\beta_2 = 0.33051, \quad 68\% : [0.32308, 0.33769], \quad 95\% : [0.31526, 0.34502], \quad (86)$$

$$\beta_4 = 0.04484, \quad 68\% : [0.03676, 0.05288], \quad 95\% : [0.02809, 0.06113]. \quad (87)$$

The posterior correlation is $\rho(\beta_2, \beta_4) = -0.835$. If the quoted experimental errors are used without a scale parameter, the 68% intervals contract to $[0.32826, 0.33259]$ and $[0.04258, 0.04728]$, respectively. Because the emulator fit has $\chi_\nu^2 = 11.22$, these statistical-only intervals should not be interpreted as total physical uncertainties.

At Eq. (85), the emulator and standalone Numerov results are especially consistent: the median relative error is 0.164%, the maximum is 0.702%, and the latter occurs at 54.2261 MeV. The emulator gives $\chi^2 = 415.27$ and $\chi_\nu^2 = 11.22$; exact Numerov at the same point gives $\chi^2 = 423.39$ and $\chi_\nu^2 = 11.44$. The cross section, barrier distribution, and joint posterior are shown in Figs. 6–8.

4.4.4 Interpretation of the two tests

Allowing γ to vary lowers the emulator minimum from $\chi^2 = 415.27$ to 376.60. However, exact Numerov evaluated at the two emulator-selected optima gives 423.39 for the axial point and 430.08 for the three-dimensional point. The current exact check therefore does not confirm that the finite- γ emulator optimum improves the high-fidelity fit. The axial emulator is presently more reliable locally, as shown by its 0.702% maximum error. The three-dimensional study should be repeated with an enlarged γ interval and extra exact training points near the upper boundary before drawing a structural conclusion about triaxiality.

In both cases the remaining large data residual is a property of the fixed physical model and potential choice, rather than an online-emulator error. The barrier distribution further differentiates these questions because the second derivative amplifies both experimental noise and small interpolation errors. It should be reported alongside the cross section, but not silently combined with the cross-section likelihood.

4.5 Current workbench flow

Mode 1 exposes Standard Numerov and Octupole + triaxial γ Numerov. A standard run also produces the Hagino same-input baseline. Mode 2 has no physical-model selector: its reference is always the triaxial source. The user first sets the reaction, radii, couplings, active coordinates, bounds, and custom or adaptive training design. After training, the workbench shows the allowed domain, performs rapid multi-parameter predictions, optionally loads experimental fusion data, and optionally generates the exact triaxial Numerov curve at the selected point.

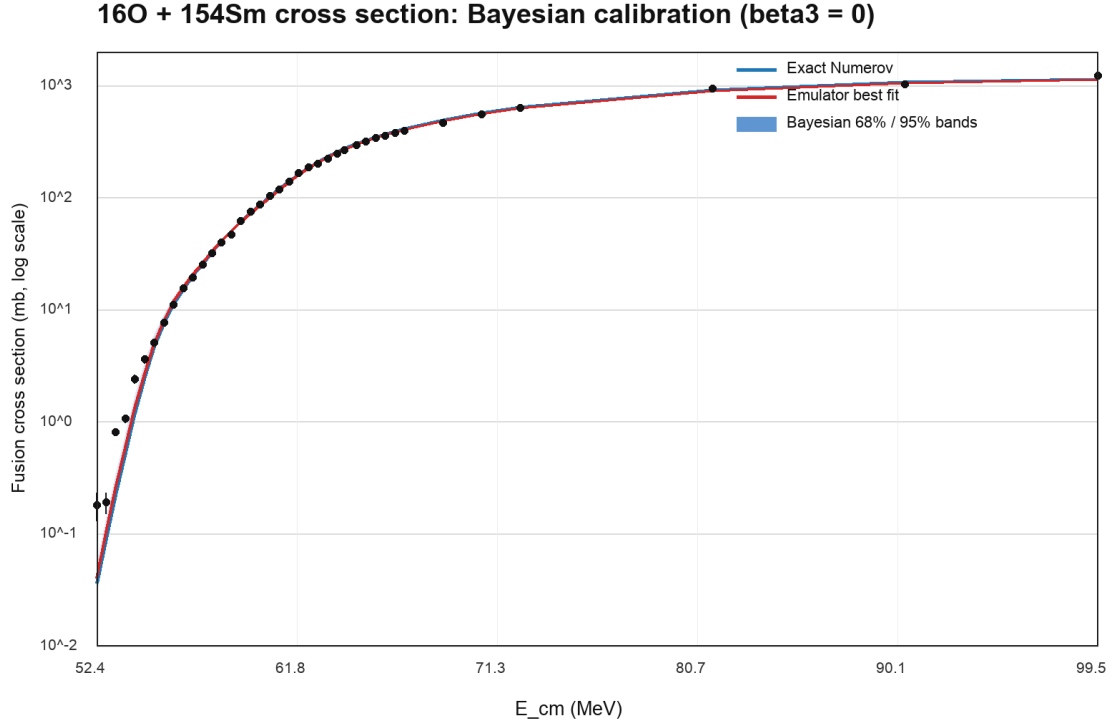


Figure 4: Three-dimensional $^{16}\text{O} + ^{154}\text{Sm}$ calibration. Black points are all 39 measured cross sections, red is the emulator maximum-likelihood curve, blue is standalone triaxial Numerov at the same parameter vector, and the shaded regions are the 68% and 95% scale-marginalized posterior predictive bands.

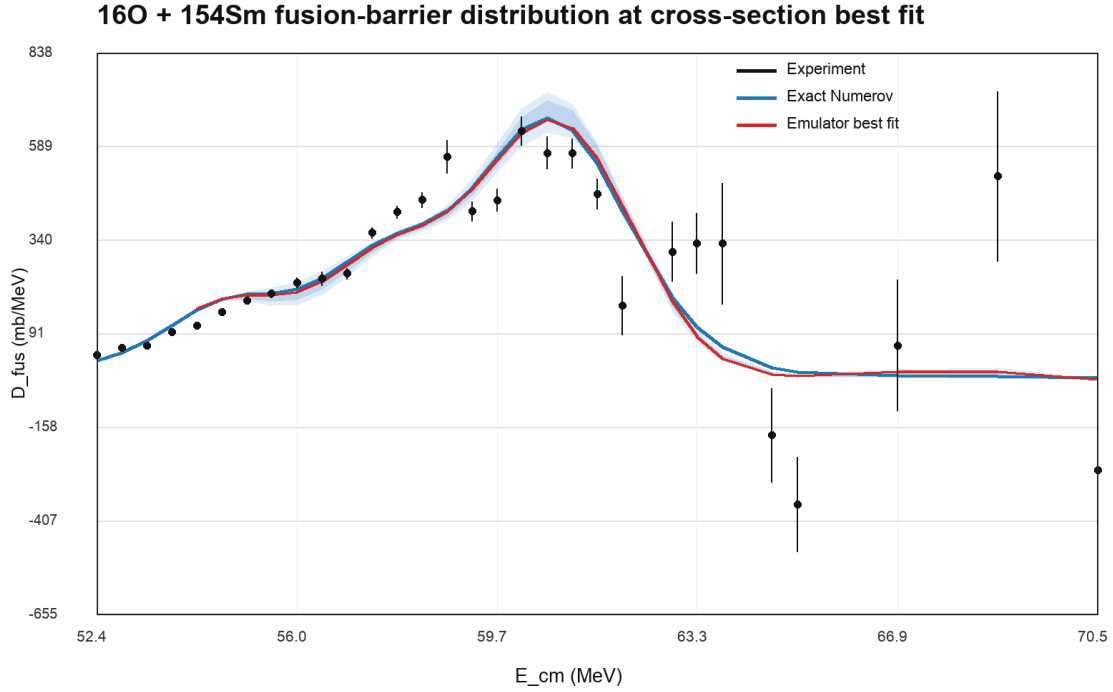


Figure 5: Fusion-barrier distribution at the three-dimensional cross-section optimum. The barrier data are not included in the likelihood; this is an out-of-fit observable check using Eq. (80).

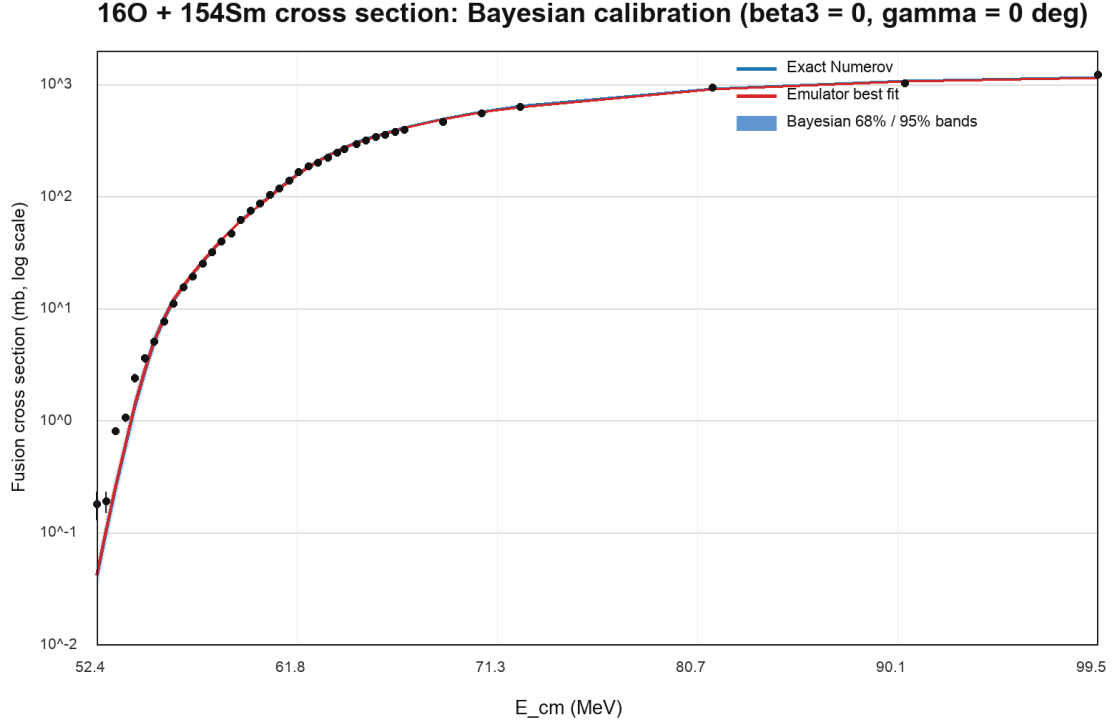


Figure 6: Two-dimensional $^{16}\text{O} + ^{154}\text{Sm}$ calibration with $\beta_3 = \gamma = 0$. The line and band definitions are the same as in Fig. 4.

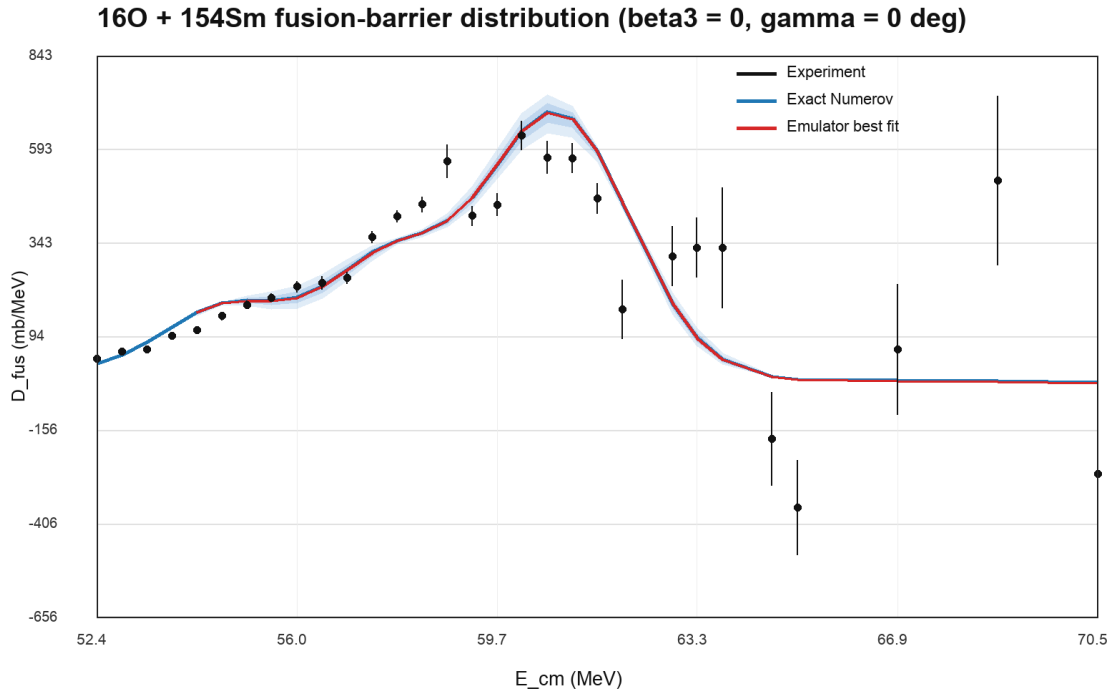


Figure 7: Fusion-barrier distribution at the two-dimensional cross-section optimum, evaluated with $\Delta E = 1.8$ MeV.

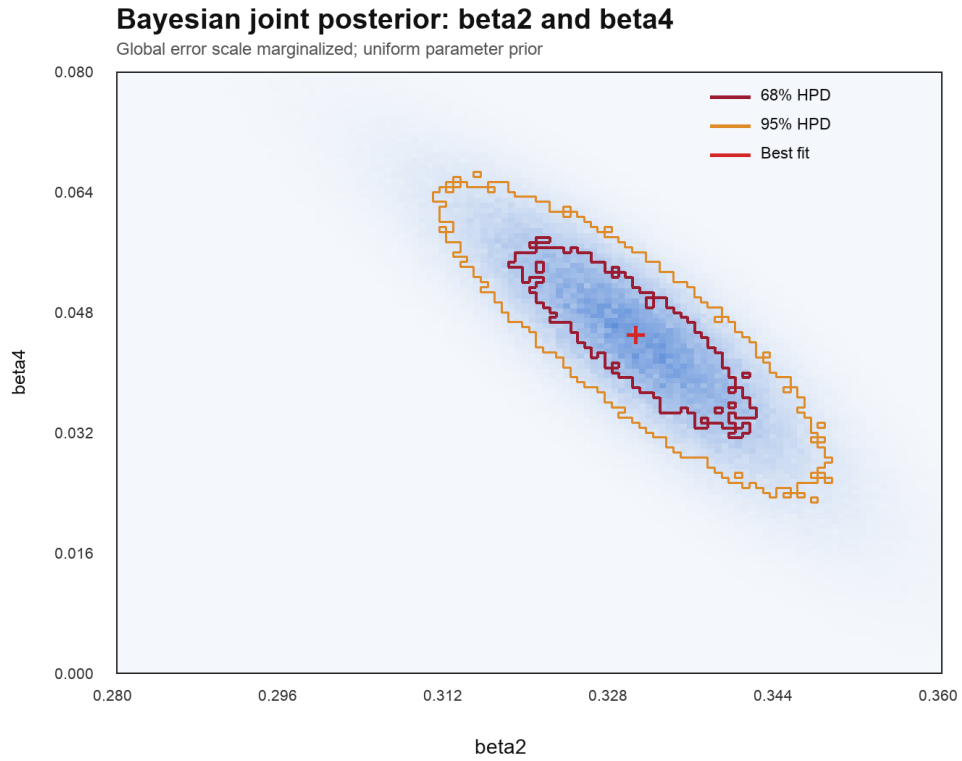


Figure 8: Scale-marginalized joint posterior for (β_2, β_4) at $\gamma = 0$. The red cross is the maximum-likelihood point; dark-red and orange contours enclose the 68% and 95% highest-posterior-density regions.

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